P0AA6-F1、 P0AA6-F0、 P0AA7-00、 P0AA8-F0、 P0AA8-F1、 P0AA9-00

Fault diagnosis code
P0AA6-F1：High voltage relay closed, isolation level 2 fault
P0AA6-F0：High voltage relay closed, isolation level 1 fault
P0AA7-00：isolation detection circuit fault
P0AA8-F0：High voltage relay opened, isolation level 2 fault
P0AA8-F1：High voltage relay opened, isolation level 1 fault
P0AA9-00：double isolation resistance is low
Detection tools
1 Multimeter, diagnostic apparatus and tramegger.
Possible causes
• BMS fault • Cooling system fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P0AA6-F1
The insulation value of the high voltage system is lower than the fault threshold: insulation resistance value ⩽ 500Ω/V(DC charging ⩽ 100Ω/V)(when high voltage relay closed)
P0AA6-F0
The insulation value of the high voltage system is lower than the fault threshold: insulation resistance value ⩽ 750Ω/V(DC charging ⩽ 500Ω/V)(when high voltage relay closed)
P0AA7-00
When the two Mos tubes are closed, the positive voltage and negative voltage do not conform to the voltage: The sum of voltage values ⩽ 0.9*Pack voltage The sum of voltage values （Vp,Vn) ⩽ 1.1*Pack voltage Invalid insulation sampling voltage
P0AA8-F0
The high voltage insulation values between main positive relay and main negative relay are lower than fault threshold. Insulation resistance value ⩽ 500Ω/V (DC charging ⩽ 100Ω/V） (when the high voltage relay closed)
P0AA8-F1
The high voltage insulation values between main positive relay and main negative relay are lower than alarm threshold: Insulation resistance value ⩽ 750Ω/V (DC charging ⩽ 500Ω/V） (when the high voltage relay closed)
P0AA9-00
Positive insulation and negative insulation value are below than alarm threshold value: insulation resistance value ⩽ 500Ω/V (DC charging ⩽ 100Ω/V）

To set the effect of a fault code condition
High voltage component does not work properly

Troubleshooting steps		
Step 1	General inspection.	
		1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO ➞ Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 If there is any fault code except for P0AA6-F1、P0AA6-F0、P0AA7-00、P0AA8-F0、P0AA8-F1、P0AA9-00?
		YES ➞ Refer to:DTC Summary list(BMS).
NO ↓		
Step 3	Check the connecting cable of BMS.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
		NO ➞ Repair or replace the faulted parts.
YES ↓		

Step 4	Check the insulation of BMS.
	1 Use a megohm meter to check the insulation resistance of the BMS housing and grounding according to the procedure in the maintenance manual. **Standard value（insulation level 1）＞ 750Ω/V （DC charging ＞ 500Ω/V）** **Standard value（insulation level 2）＞ 500Ω/V （DC charging ＞ 100Ω/V）** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the insulation of BMS high-voltage cables.
	1 Use a megohm meter to check the insulation resistance of the BMS high voltage and grounding according to the procedure in the maintenance manual. **Standard value（insulation level 1）＞ 750Ω/V （DC charging ＞ 500Ω/V）** **Standard value（insulation level 2）＞ 500Ω/V （DC charging ＞ 100Ω/V）** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Perform initialization.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
	NO → Repair or change the BMS module.
YES ↓	

Step 7	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1000_1 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	